



DETRAINING



The dreaded loss of the gains from so many hours of training (when a holiday, an injury or a busy period at work keeps you from training for a while), in other words detraining, comes down to the reversibility principle. Unfortunately, there's still no research into detraining in climbing but there are a number of studies on detraining strength and endurance in other sports which we can use for reference.

As a starting point, we know that the more extreme the adaptation, the faster it's lost. Usain Bolt has only once run the 100 metres in 9.58 seconds. If he had run the following day, he would probably have got a slower time. Adam Ondra made the first ascent of *Silence* (F9c) on 3 September 2017, after four years of training and trying the route. There's very little chance that he could have climbed it again the following day.

Let's look at some of the research into detraining:

- There's a 23 per cent reduction in the activity of mitochondria and oxidative enzymes after three weeks of inactivity, dropping by a further 23 per cent from weeks four to eight (Mujika, 2012).
- There's a 22 per cent increase in postprandial insulin (the hormone that makes us put on weight, which 'stores calories' after a meal) after 6.5 days of inactivity (Mujika and Padilla, 2003).
- There's an 86 per cent increase in lipoprotein lipase activity in adipose tissue (the storage of fatty acids in fatty tissue, storing fuel in subcutaneous fat) after just 14 days of inactivity. In contrast, there's a 75 per cent reduction in lipoprotein lipase activity in muscle (allowing us to burn these fatty acids) during the same 14-day period (Mujika and Padilla, 2003). In other words, a shift in enzyme activity meaning more fat is stored and less fat is burnt.
- A study by Alomari *et al.* (2010) examined the effects of handgrip training and detraining on vascular resistance. Participants did 20 minutes of rhythmic handgrip exercises, at 60 per cent of their MVC (maximum voluntary contraction), five days a week for a total of four weeks. After these four weeks of training, forearm blood flow had increased significantly but just one week after they stopped training, blood flow had returned to pre-training levels.
- There's a 39 per cent reduction in muscle glycogen stores after seven days of inactivity (Mujika and Padilla, 2003).
- A study on a group of swimmers who swam the 200 metres freestyle at 90 per cent of their best time for that distance examined levels of blood lactate and HCO_3 (bicarbonate, which is the main buffering system for lactate: it's the 'broom' that can 'sweep away' this metabolic waste) over four weeks of inactivity:

	Weeks without training			
	0	1	2	4
Lactate (mmol/L)	4.2	6.3	6.8	9.7
HCO_3 (mmol/L)	21.2	19.5	16.1	16.3

Costill *et al.* (1985)

The table shows a dramatic rise in blood lactate levels (the swimmers lost oxidative capacity in favour of anaerobic lactic metabolism) and significantly lower levels of HCO_3 (an impaired ability to flush out metabolic waste) after just four weeks of inactivity.

- Luckily, similar studies on detraining strength are a little more encouraging. Kraemer *et al.* (2002) studied the loss of maximum strength and power in a group of men involved in recreational strength training (not at elite level). After six weeks of inactivity, the group experienced an almost negligible loss of maximum strength and around 10 per cent loss of power.
- García-Pallarés *et al.* (2010) studied the effects of five weeks of inactivity on world-class kayakers. Five weeks after they stopped training, the participants had lost around 9.8 per cent on their 1RM bench press.

This gives a good indication of what happens when we stop training. A house can take years to build but be demolished in a matter of minutes. The same can be said of physical fitness and performance, especially in relation to endurance. Use this chapter as motivation to never give up on training. Use this book to understand that to perform at your best, in most cases it's counter productive to train at your absolute limit.

No pain, more gain. Don't train more: train better.